

Jiwon Park



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SUMMARY

Master's student interested in improving robotic perception and scene understanding through deep learning-based sensor data processing, with applications to robust autonomous robot navigation.

RESEARCH INTERESTS

- Generative Perception for Robotics
- Robust Autonomous Navigation
- Planning on Neural Maps

EDUCATION

Mar. 2025 - Present	Korea Advanced Institute of Science and Technology (KAIST) M.S course in Robotics Program	Daejeon, Korea
Mar. 2019 - Aug. 2024	Kyung Hee University B.S course in Department of Electronic Engineering, Software Convergence (Robot Vision Track)	Yongin-si, Korea

PUBLICATIONS

INTERNATIONAL CONFERENCES

May. 2026	DreamFlow : Local Navigation Beyond Observation via Conditional Flow Matching in the Latent Space IEEE International Conference on Robotics & Automation (ICRA), 2026 Jiwon Park* , Dongkyu Lee*, I Made Aswin Nahrendra, Jaeyoung Lim and Hyun Myung
Dec. 2023	BumpyPatch: Heightmap-based Outdoor Point Cloud Segmentation to Find Less Bumpy Road IEEE International Conference on Robotic Computing (IRC), 2023 Jiwon Park and Hyoseok Hwang
Dec. 2022	Outdoor Visual SLAM and Path Planning for Mobile-Robot IEEE International Conference on Robotic Computing (IRC), CHARMS workshop, 2022 Seongil Heo, Jueun Mun, Jiwoong Choi, Jiwon Park , Eric T. Matson

DOMESTIC JOURNAL

Aug. 2024	Transformer based Collision Detection Approach by Torque Estimation using Joint Information Journal of Korea Robotics Society Vol.19 No.3 Jiwon Park , Daegyu Lim, Sumin Park, Hyeonjun Park
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EXPERIENCE

Mar. 2024 - Sep. 2024	NAVER LABS Internship • Developed software of inference board for robot, enhancing on-board processing capabilities • Implemented deep learning model optimization techniques for low-cost NPUs • Integrated third-party robots into multi-robot intelligence system (ARC) • Developed monitoring applications for robot experiments	Seongnam-si, Korea
Dec. 2023 - Feb. 2024	ROBROS Internship • Developed experimental environments for robotic arms using MuJoCo simulation platform • Proposed a transformer-based model for detecting collisions in robotic arms without external sensors, relying on joint position and torque controllers	Seoul, Korea

Apr. 2022 - Aug. 2022	Purdue University Student Intern	West Lafayette, IN, USA
	• Conducted research on outdoor mobile visual SLAM (Simultaneous Localization and Mapping) • Focused on enhancing accuracy and reliability of visual navigation for mobile robots	

Projects

Mar. 2025 - Present	Development of mobile intelligence SW for autonomous navigation of legged robots in dynamic and atypical environments for real application	KEIT
	• Implemented multi-robot autonomous navigation and path planning for agricultural field robots in real-world farm environments	
Mar. 2025 - Present	Dexterous and Agile Humanoid Robotics for Industrial Applications (DARIUS)	KEIT
	• Developed uncertainty- and semantic-aware path planning for humanoid robot navigation in indoor industrial environments	
Sep. 2025 - Present	Development of robot technology for forest fire response in irregular mountainous areas	KIRO
	• Developed a target-following navigation algorithm using future trajectory prediction in dynamic and unstructured environment	

AWARD & HONORS

Mar. 2025	2nd place in Quadruped Robot Challenges (QRC), ICRA • IEEE Robotics and Automation Society
Mar. 2023 - Aug. 2024	Full-ride scholarship until graduation • KC Future Foundation
Apr.2022- Aug.2022	Purdue-Korea Software Square Program • Ministry of Science and ICT
Dec.2022	SW Festival, Grand Prize • Kyung Hee University, Software College
Nov.2021	The World Embedded Software Contest, Participation Award • Korea Embedded Software and System Industry Association
Dec.2020	Open SW Utilization Contest, Excellence Prize • Kyung Hee University, Software College
Jul.2020	Autonomous Mini-drone Aviation Competition, Excellence Prize • Korean Institute of Electrical Engineers

LANGUAGES

English	Proficient (TOEIC : 945)
Korean	Native